SECTION–I : (i) (Maximum Marks : 42)

This section contains **14 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. In which of the following reactions, Na_2O is not formed as a product?

(A) $Na_2O_2 + Na \rightarrow$ (B) $NaNO_2 + Na \rightarrow$

(C) $Na_2O_2 + SO_2 \rightarrow$ (D) $NaNO_3 + Na \rightarrow$

2. By product of 'Solvay process' is:

(A) $CaCl_2$ (B) NH_4Cl (C) CO_2 (D) NH_3

3. Which of the reaction is **INCORRECTLY** written with NH_3 in different condition (Liq. NH_3; gas NH_3 ; NH_3 (aq.) solution)

(A) $Na + NH_3(\ell) \longrightarrow Na^+(NH_3)_x + e^-(NH_3)_y$.

(B) $Pb^{2+} + 4NH_3aq \longrightarrow [Pb(NH_3)_4]^{2+}$

(C) $2Na + 2NH_3(g) \longrightarrow 2NaNH_2 + H_2(gas) \uparrow$

(D) $AgNO_3 + \underset{\text{(excess)}}{2NH_4OH}$ (aq. solution of NH_3) $\rightarrow [Ag(NH_3)_2]^+ + NO_3^- + 2H_2O$

4. Which of the following ion is unstable in aqueous solution?

(A) Co^{+2} (B) Fe^{+2} (C) Mn^{+2} (D) Cu^{+1}

5. Slaked lime is obtained when water is added to :

(A) $CaSO_4.\frac{1}{2}H_2O$ (B) $CaCl_2$ (C) CaO (D) $CaCO_3$

6. Choose **CORRECT** statement

(A) In $(NH_4)_2Cr_2O_7$ geometry at Nitrogen & each Chromium is NOT same

(B) On heating $(NH_4)_2\ Cr_2O_7$ give N_2 as a product and it is endothermic reaction.

(C) All Nitrate salt on decomposition gives atleast one gaseous product

(D) Metal nitrates are generally insoluble in water.

7. Metal hydroxide $\xrightarrow[\text{passed}]{CO_2 \text{ is}}$ Metal carbonate $\xrightarrow[\text{is passed}]{\text{excess } CO_2}$ metal bicarbonate

Which of the following metal hydroxide produce soluble carbonate as well as soluble bicarbonate in the above reaction sequence.

(A) $Ca(OH)_2$ (B) NH_4OH

(C) $Ba(OH)_2$ (D) KOH

8. $BeCl_2 + LiAlH_4 \longrightarrow X + LiCl + AlCl_3$

Total number of $3c–4e^{\ominus}$ bond present in monomer form of compound (X)

(A) 1 (B) 2

(C) 3 (D) 0

9. $KO_2 + H_2O \longrightarrow KOH + O_2 +$ product (X)

Choose the correct about product (X)

(A) It has open book like structure (B) It stabilized by urea

(C) It decomposes in presence of light (D) All of these

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10. Which of the following metal does not react with nitrogen and cannot form nitride
(A) Al (B) Mg (C) Cs (D) Li

11. Consider the reaction

$K_2[Ni(CN)_4] + NH_3 + 2K(\text{potassium}) \longrightarrow$ complex Ⓐ

complex Ⓐ gives five ions in aqueous solution then choose **CORRECT** statements.

(A) complex is sp^3 hybridised & diamagnetic
(B) complex is sp^3 hybridised and paramagnetic
(C) complex is dsp^2 hybridised and diamagnetic
(D) In complex $\overset{\ominus}{C}N$ ligands are replaced by NH_3 ligands

12. The electronic configuration of central atom of complex $[MnO_4]^{2-}$ is
(A) $t_2^1\, e^0$ (B) $t_2^0\, e^0$ (C) $e^0\, t_2^0$ (D) $e^1\, t_2^0$

13. On heating 'Trona ($Na_2CO_3.NaHCO_3.2H_2O$)' with the help of Bunsen Burner, we get _____ as solid residue
(A) Na_2CO_3 (B) Na_2O
(C) $Na_2CO_3.H_2O$ (D) $NaHCO_3$

14. Gypsum is added into the Portland Cement to
(A) Stop setting process (B) Slow down the setting process
(C) Fasten the setting process (D) Adjust the viscosity of the slurry

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SECTION-I(ii) : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

15. Which of the following statement(s) is/are **CORRECT**.
(A) Among alkaline earth metal oxalate, MgC_2O_4 is least soluble in water
(B) Among alkali metal carbonate Li_2CO_3 is least stable.
(C) KO_3 has paramagnetic and orange red colour
(D) KO_2 is used in ill ventilated room

16. Which of the following reactions give correct product?

(A) $Mn^{2+} + (NH_4)_2S_2O_8 + H_2O \longrightarrow MnO_4^{2-} + SO_4^{2-} + NH_4^+ + H^+$

(B) $K^+ + NO_2^- + Co^{2+} \longrightarrow K_4[Co(NO_2)_6] + NO + H_2O$

(C) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

(D) $Cu + 2H_2SO_4$ (conc.) $\longrightarrow CuSO_4 + SO_2 + 2H_2O$

17. Choose the **INCORRECT** order of the indicated property given in the options.

(A) He < Ne < Kr < Ar : Clathrate compound formation tendency order

(B) $NaO_2 < KO_2 < RbO_2 < CsO_2$: Thermal stability order

(C) $BeF_2 < BaF_2$: Solubility order

(D) LiF > LiCl > LiI : Melting point order

18. Which of the properties are **INCORRECT** regarding $NaHCO_3$ and $KHCO_3$.

(A) Both are fairly soluble in water

(B) Both have similar hydrogen bonded structure of HCO_3^- ion in their solid

(C) Both solid consists of ion-dipole type of interaction

(D) Both on heating produces respective carbonate as residue

19. Which of the following is/are **NOT CORRECT** for Lithium –

(A) It's hydrogen carbonate exist in solid form

(B) It's carbonate is most stable to heat amongest the group I-A metals

(C) It's main combustion product in excess oxygen is LiO_2

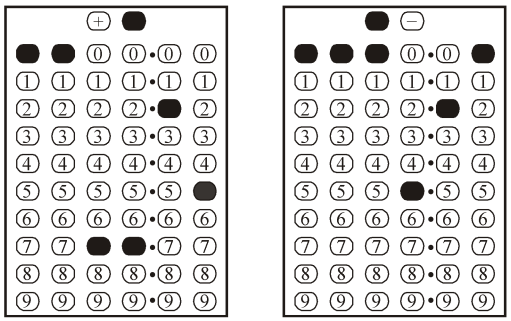
(D) It's chloride is soluble in pyridine

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SECTION-II : (Maximum Marks: 6)

- This section contains **TWO** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. How many reactions give **CORRECT** products ?

(i) $2Na + NH_3\ (liq) \rightarrow Na_2NH + H_2\uparrow$

(ii) $Na + (x+y)\ NH_3\ (g) \rightarrow [Na(NH_3)_x]^{+1} + [e(NH_3)_y]^{-1}$

(iii) $Ni^{2+} + 4NH_3(excess) \rightarrow$ $[Ni(NH_3)_4]^{2+}$

Blue solution

(iv) $BaCl_2 + K_2CrO_4 \rightarrow$ $Ba^{2+} + CrO_4^{2-} + 2K^+ + Cl^-$

soluble products

(v) $NH_4Cl + NaNO_2 \xrightarrow{\Delta} NaCl + N_2O + H_2O$

(vi) $Hg_2Cl_2 + 2NH_3 \rightarrow Hg + Hg\begin{<}{NH_2}{Cl}\ + NH_4Cl$

Nice black ppt

2. Find the sum of the oxidation state of S-atoms present in compounds of Lithophone.

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